POLKA: Towards Leakage-Resistant Post-Quantum CCA-Secure Public Key Encryption

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Post-quantum cryptography (PQC) and Side-Channel Attacks (SCA)

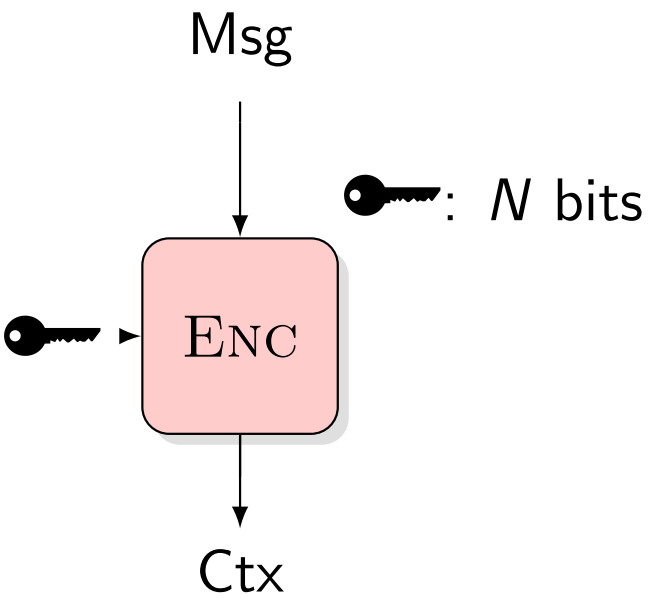


Side-channel attacks:

- Generic against lattices: [RRCB19], [NWDP22], [KAPFA21],
- Against the Number Theoretic Transform (NTT): [PPM17], [PP19], [LZHLT22]
- Against the Fujisaki-Okamoto(FO) transform: [UXTITH22]

Countermeasures are expensive: [RPBC20], [PessI16], [NWDP22], [ABHKSS22]

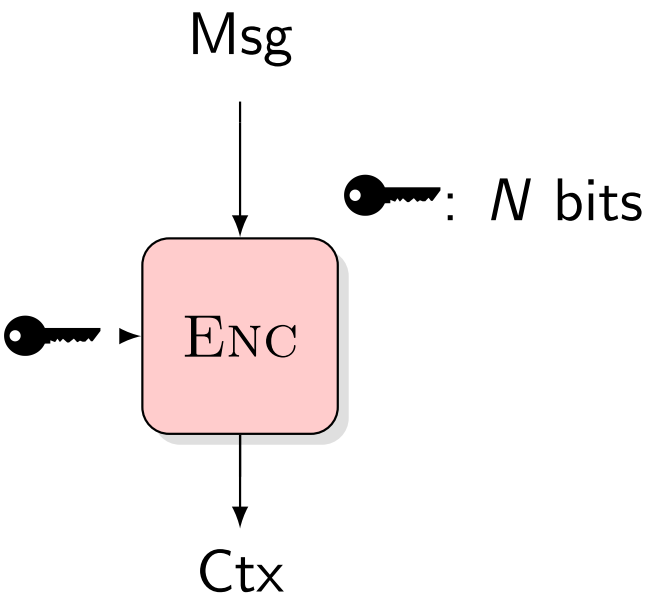
Side-channel attacks



Side-channel attacks



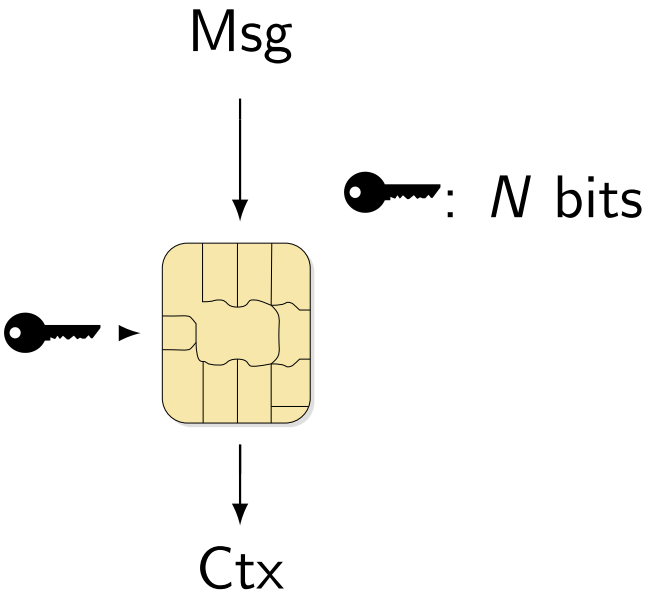
"Cryptographic algorithms don't run on paper,



Side-channel attacks

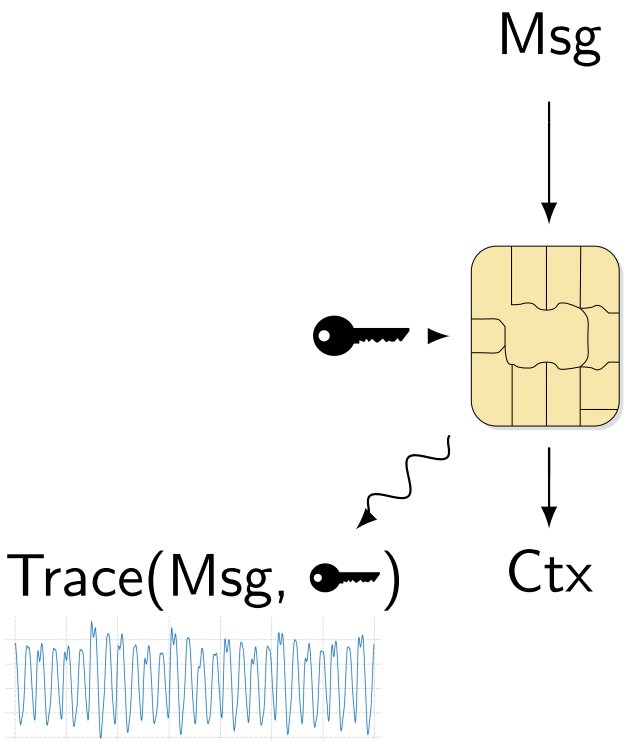


"Cryptographic algorithms don't run on paper, they run on physical devices"



Side-channel attacks

"Cryptographic algorithms don't run on paper, they run on physical devices"



For every intermediate value v, an adversary gets a noisy image of the leakage function $L_g(v)$.

Trace : power, EM, acoustics, runtime, ...

Side channel attacks - Power Analysis



SPA:

- Require only a few traces
- Can target ephemeral secret
- Typical countermeasure: parallelism, shuffling

Side channel attacks - Power Analysis



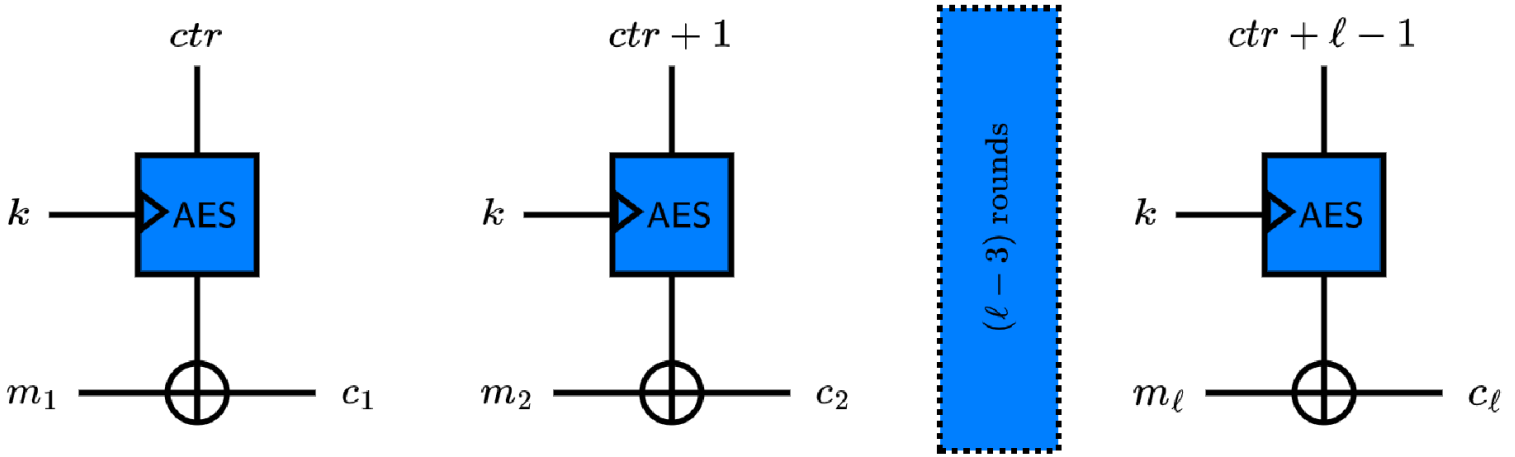
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DPA:

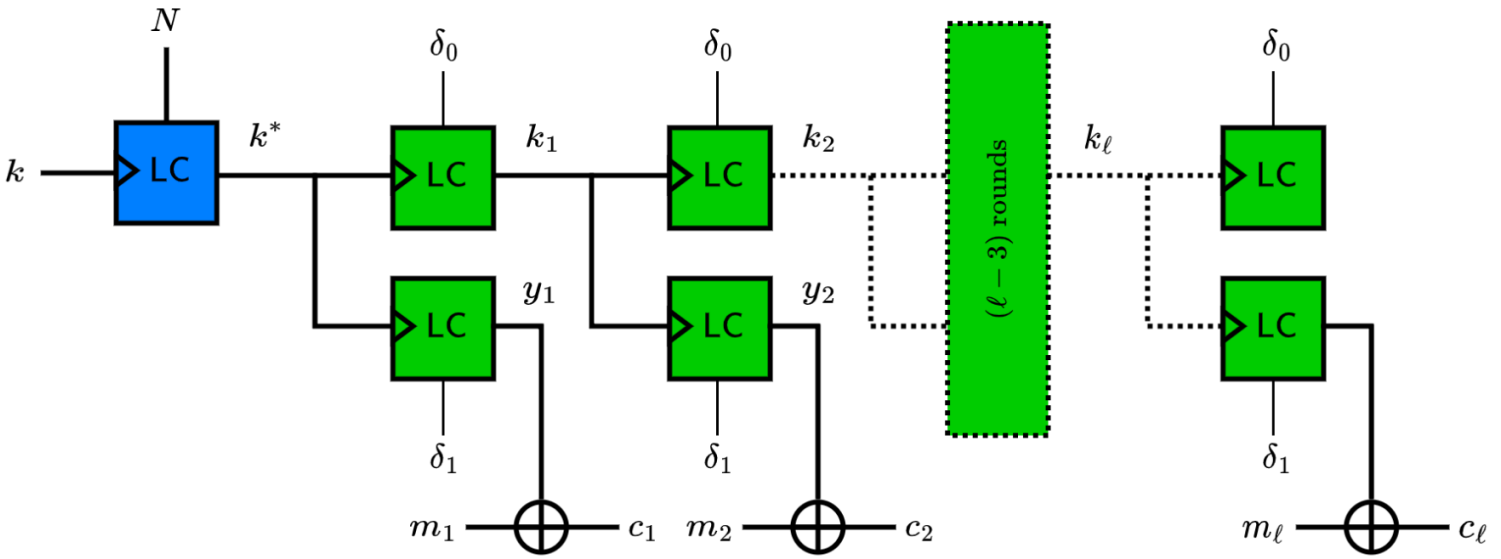
- Require a large amount of trace
- Can only target long-term secret
- Typical countermeasure: masking

Parallel with symmetric cryptography - Leveling



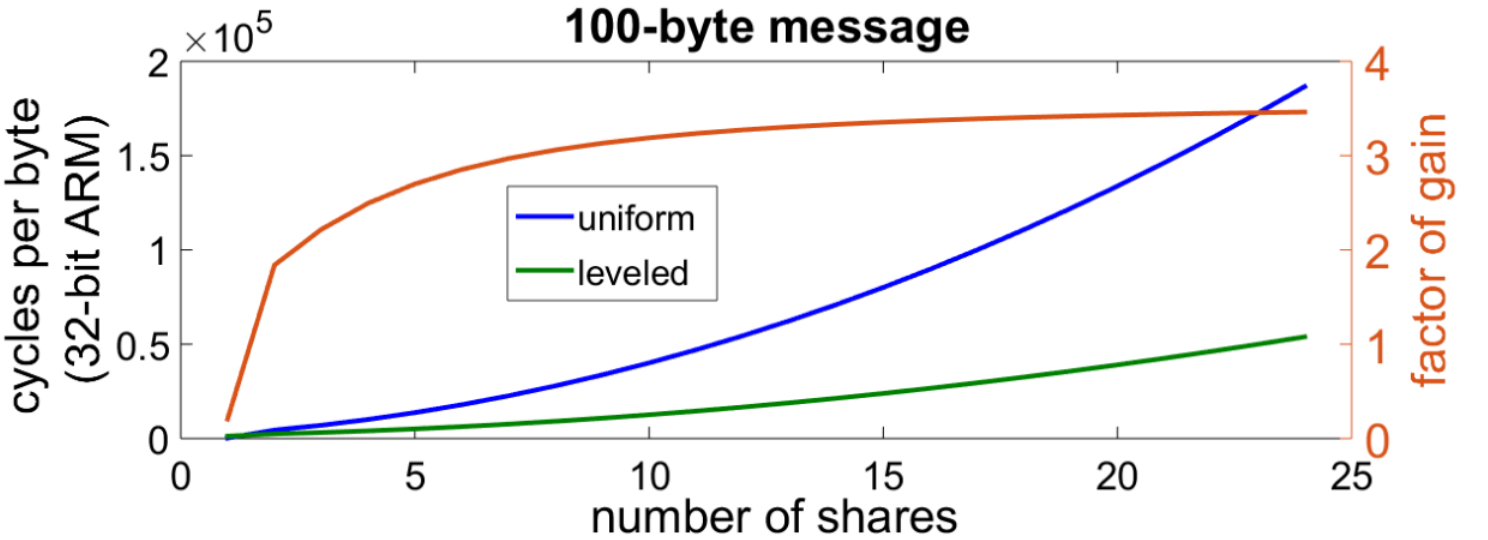
AES in CTR mode; uniformly protected against DPA

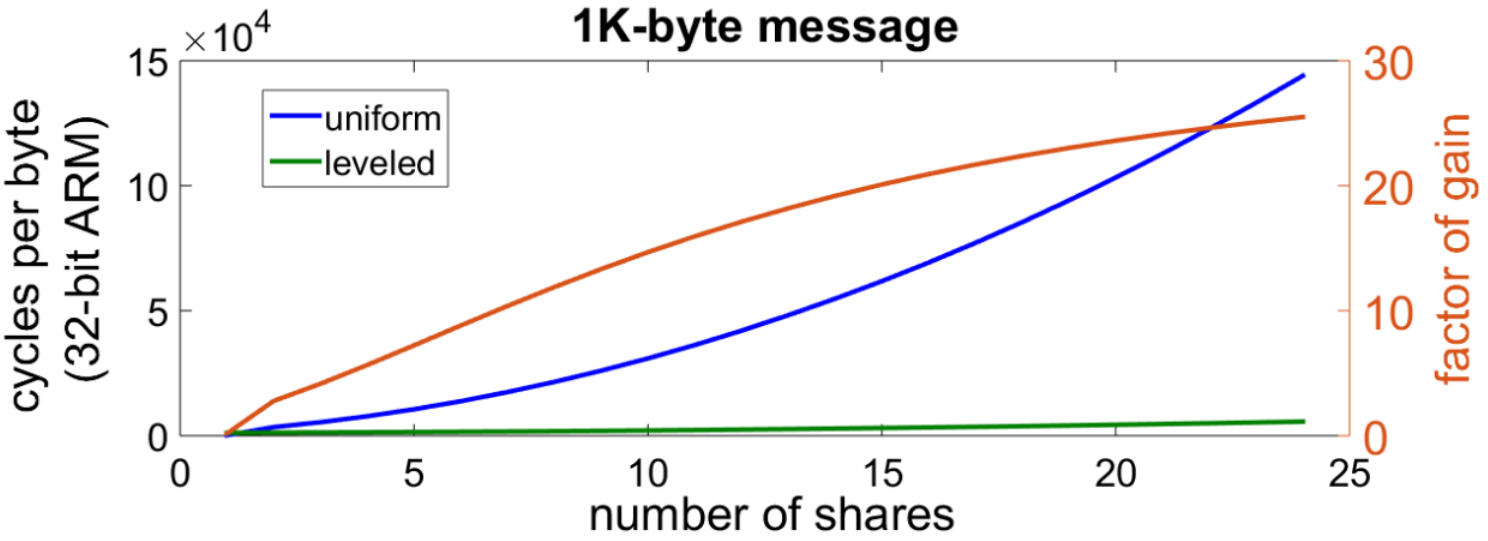
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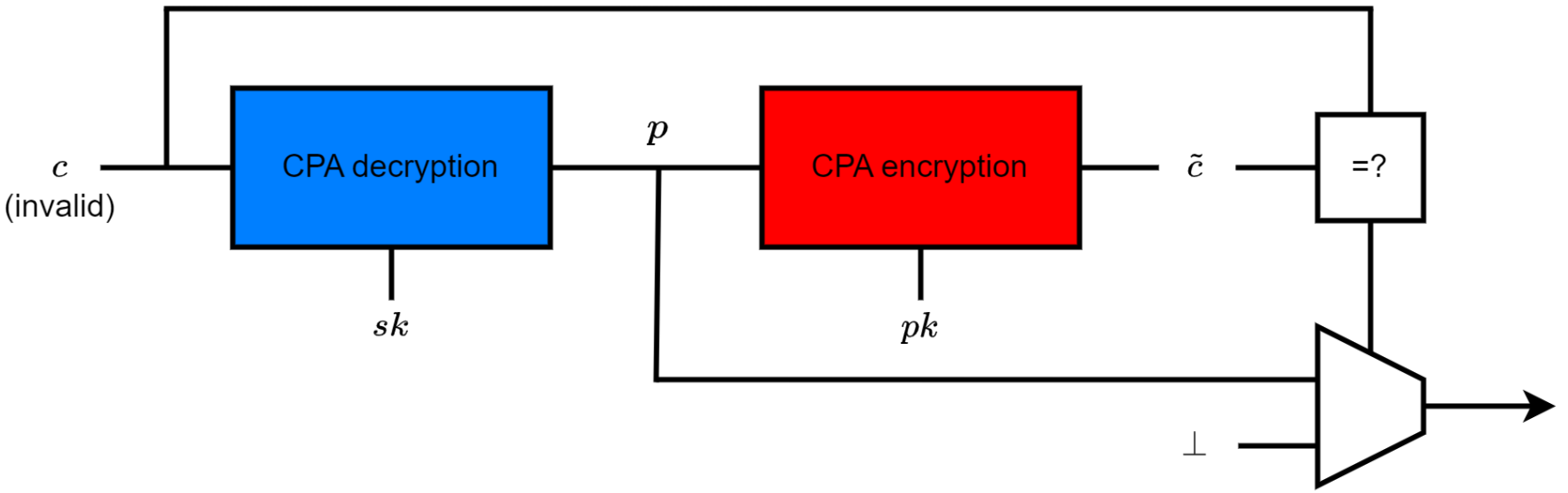
Leakage-resistant mode
Leveled implementation thanks to rekeying

Leveling impact can be massive

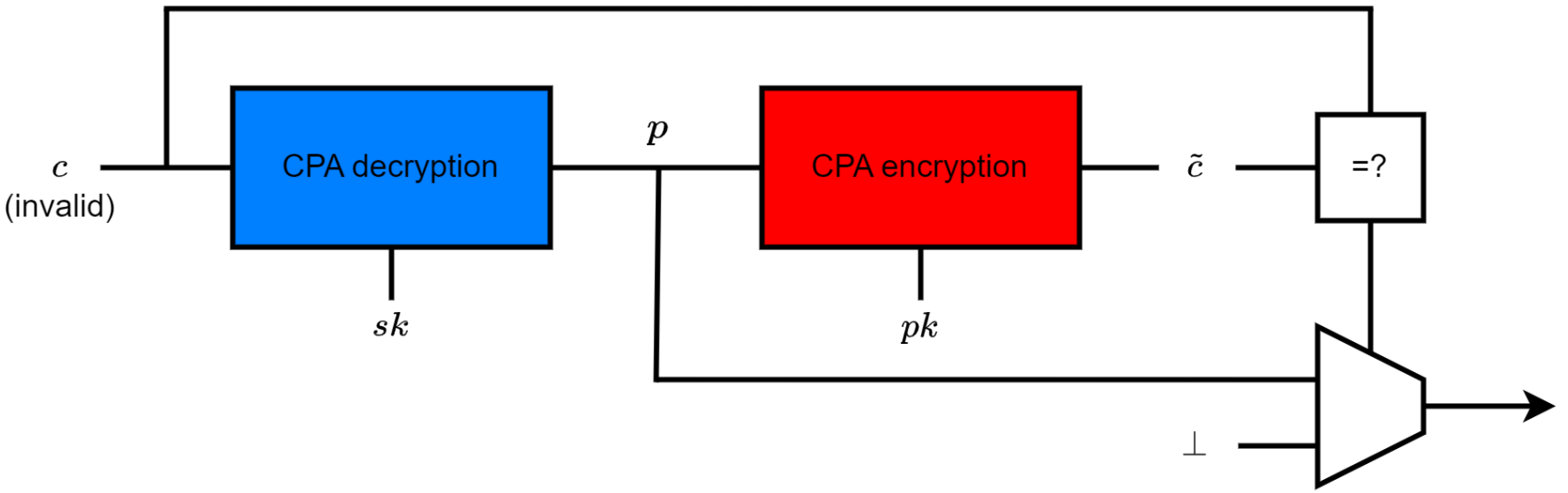




The example of FO



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Distinguishing attack:

- Simply distinguish between two values with leakage
- Even more expensive to prevent than DPA

Our contributions



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  - Proven secured in ROM & QROM

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- CCA lattice-based encryption scheme relying on **RLWE**
  - **Relatively efficient**
  - Proven secured in ROM & QROM
- Much **cheaper to protect against SCAs** thanks to tweaks:
  - CCA without FO transform
  - Dummy ciphertexts
  - Hard physical learning problems

High-level outline



Classic **[LPR10]**-like encryption scheme

$$\begin{aligned} c_1 &= a \cdot r + e_1 \\ c_2 &= b \cdot r + e_2 \end{aligned} \quad , \text{ for small } r, e_1, e_2$$

High-level outline



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$$\text{Retrieve } e_2 \Rightarrow \text{Retrieve } e_1 \text{ and } r \Rightarrow \text{Decrypt}$$

Rigidity property ([BP18])



p: intermediate modulus

$$c_2 - p \cdot c_1 \cdot sk = p \cdot (\dots) + \boldsymbol{e_2}$$
$$\boldsymbol{r} = (c_2 - e_2) \cdot b^{-1}$$
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DPA attack path

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Check $||\overline{e_1}||, ||\overline{e_2}||, ||\overline{r}|| < 2B$.

Dummy ciphertexts



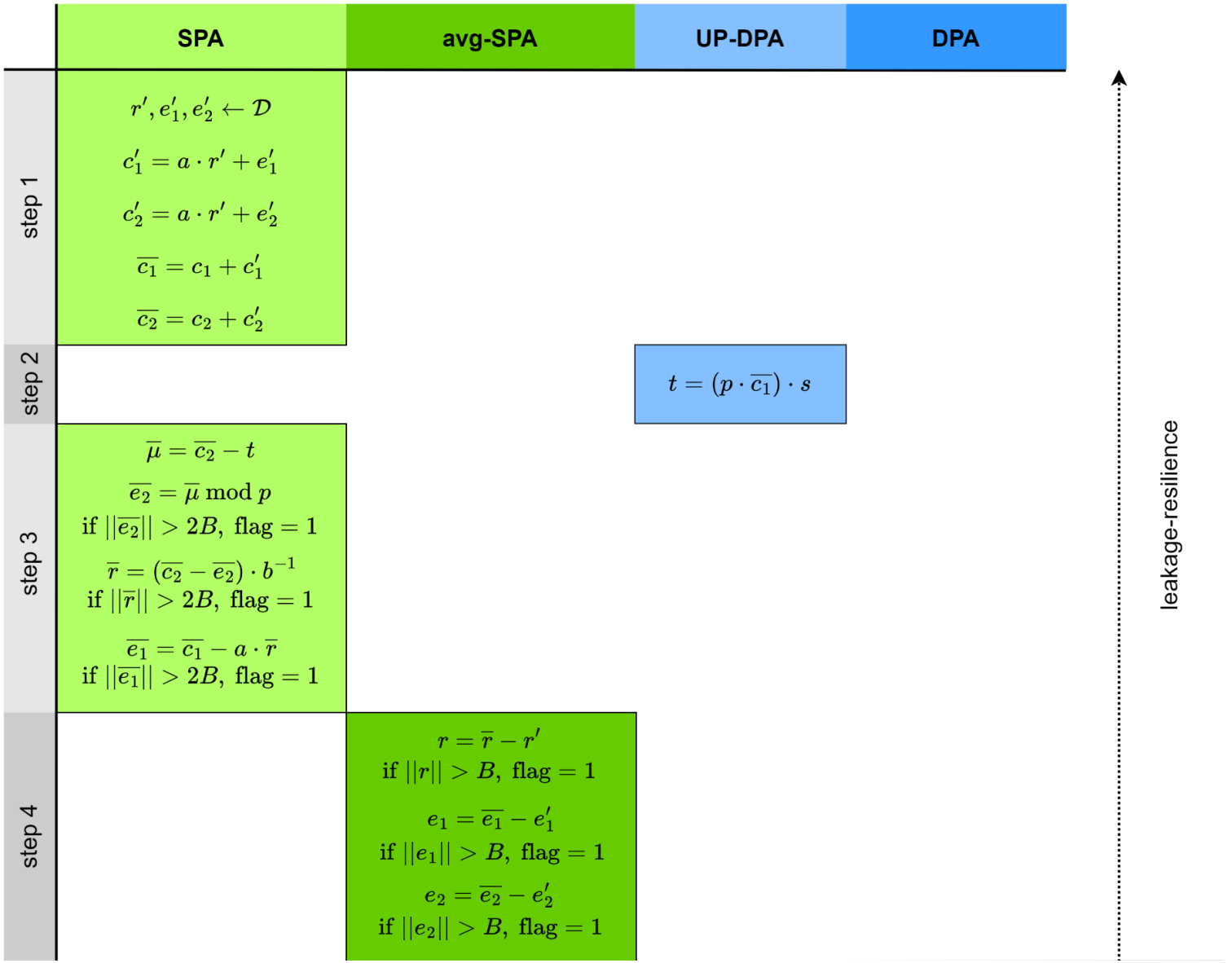
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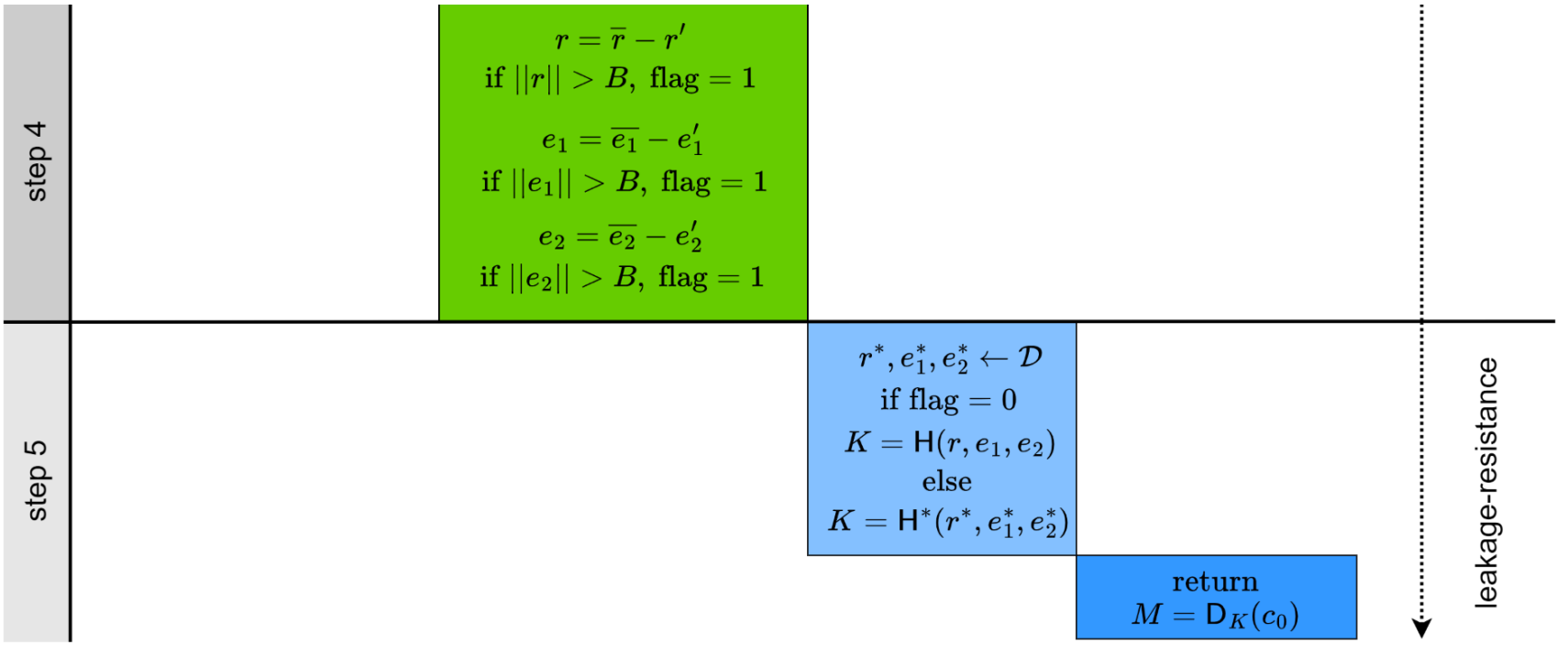
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DPA attack paths $\Rightarrow$ SPA attack paths

Leveling Polka.Dec



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Black-box security



- ROM: rigidity property ([BP18])
- QROM: reduction to implicit reduction

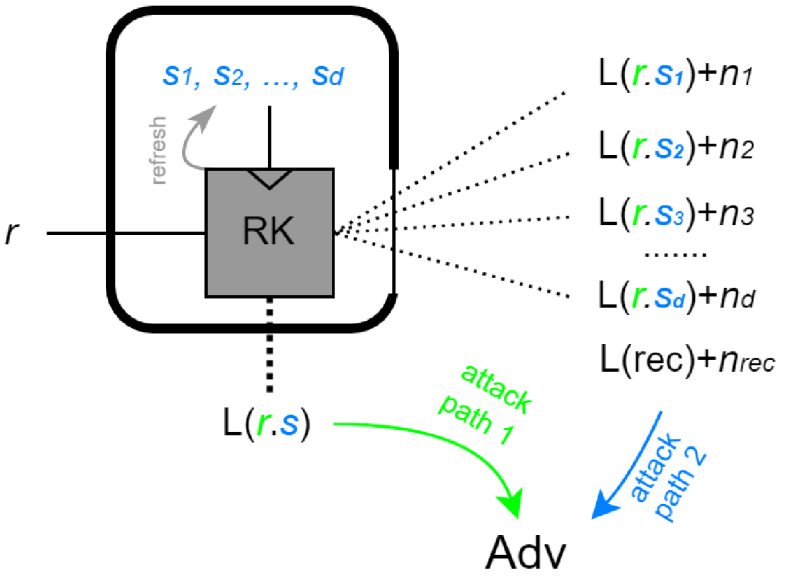
Black-box security



- ROM: rigidity property ([BP18])
- QROM: reduction to implicit reduction

Design tweaks are part of the black-box analysis

Hard physical learning problem (or free rekeying)



Rekeying with Learning With Physical Rounding (LWPR, [DMMS21])

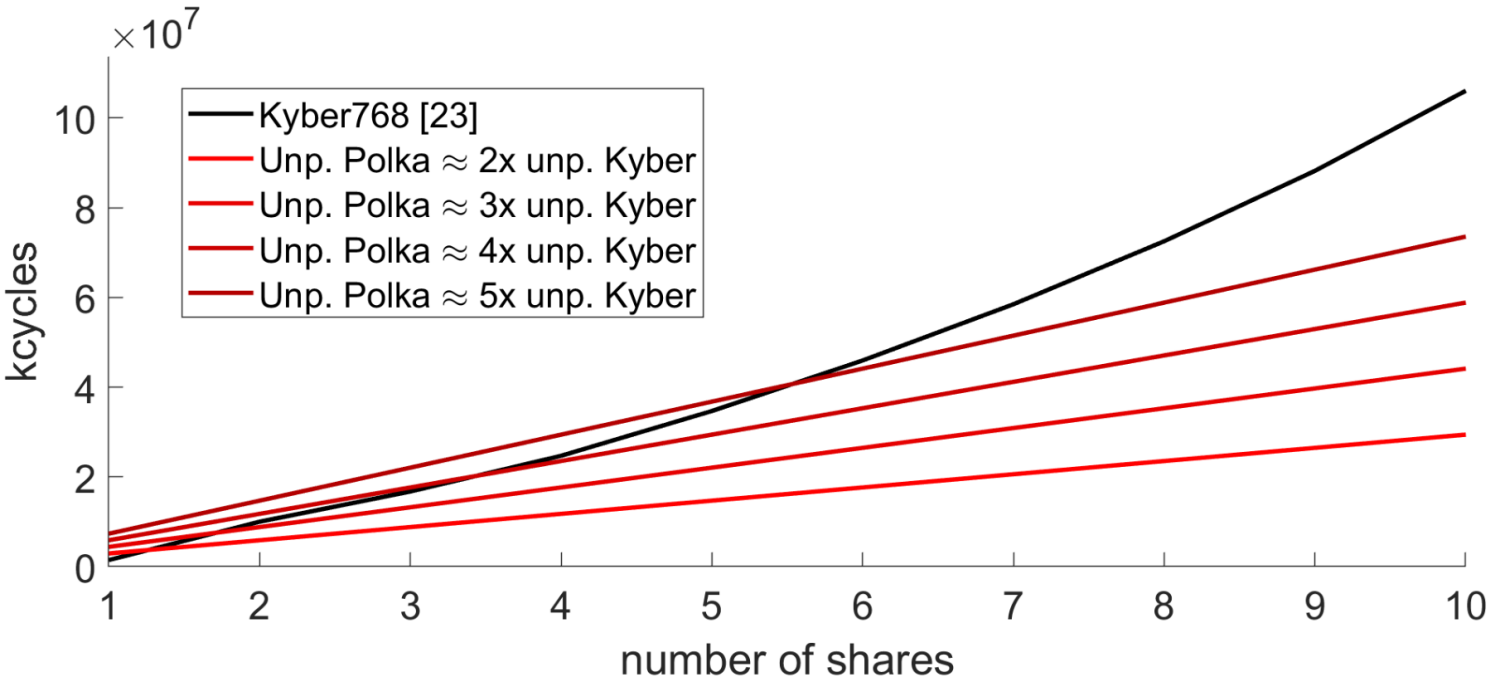
- Attack path 1 targets the ephemeral values output by the rekeying
- Attack path 2 targets the rekeying operations

Open problems



- Concrete implementation and comparison (e.g. with Kyber)
- Adapt the scheme to protect the message as well
- Proof on LWPR-like problems (reduction to LWR/LWE ?)

Conclusion



Expected time complexity of KYBER and POLKA according to orders of masking

Supplementary material - references



[RRCB19] - Generic Side-channel attacks on CCA-secure lattice-based PKE and KEMs; Prasanna Ravi, Sujoy Sinha Roy, Anupam Chattopadhyay & Shivam Bhasin

[KAPFA21] - 2Deep: Enhancing Side-Channel Attacks on Lattice-Based Key-Exchange via 2-D Deep Learning; Priyank Kashyap, Furkan Aydin, Seetal Potluri, Paul D. Franzon & Aydin Aysu

[PPM17] - Single-Trace Side-Channel Attacks on Masked Lattice-Based Encryption; Robert Primas, Peter Pessl & Stefan Mangard

[PP19] - More Practical Single-Trace Attacks on the Number Theoretic Transform; Robert Primas & Peter Pessl

[LZHLT22] - Single-Trace Side-Channel Attacks on the Toom-Cook: The Case Study of Saber; Yanbin Li, Jiajie Zhu, Yuxin Huang, Zhe Liu & Ming Tang

[RPBC20] - On Configurable SCA Countermeasures Against Single Trace Attacks for the NTT; Prasanna Ravi, Romain Poussier, Shivam Bhasin, and Anupam Chattopadhyay

Supplementary material - references

[UXTITH22] - Curse of Re-encryption: A Generic Power/EM Analysis on Post-Quantum KEMs; Rei Ueno, Keita Xagawa, Yutaro Tanaka, Akira Ito, Junko Takahashi & Naofumi Homma

[Pessl16] - Analyzing the Shuffling Side-Channel Countermeasure for Lattice-Based Signatures; Peter Pessl

[NWDP22] - Side-Channel Attacks on Lattice-Based KEMs Are Not Prevented by Higher-Order Masking; Kalle Ngo, Ruize Wang, Elena Dubrova & Nils Paulsrud

[ABHKSS22] - Systematic study of decryption and re-encryption leakage: the case of kyber; Melissa Azouaoui, Olivier Bronchain, Clément Hoffmann, Yulia Kuzovkova, Tobias Schneider & François-Xavier Standaert

[BP18] - Towards KEM Unification; Daniel J. Bernstein & Edoardo Persichetti

[DMMS21] Exploring Crypto-Physical Dark Matter and Learning with Physical Rounding; Sébastien Duval, Pierrick Méaux, Charles Momin & François-Xavier Standaert

[LPR10] On ideal lattices and learning with errors over rings; Vadim Lyubashevsky, Chris Peikert & Oded Regev

Thanks to L. Masure, FX. Standaert, O. Bronchain and C. Momin for letting me use their figures.

Supplementary material - FO transform

